1.4.3 Digital Thermometer

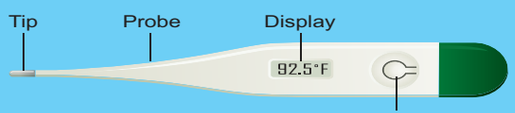
Here is a lot of concern over the use of mercury in thermometers. Mercury is a toxic substance and is very difficult to dispose of if a thermometer breaks. These days, digital thermometers are available which do not use mercury. Instead, it has a sensor which can measure the heat coming out from the body directly and from that can measure the temperature of the body.



Digital thermometers are mainly used to take the body temperature.

ACTIVITY 5

Use of Digital thermometer



1. Wash the tip with warm (not hot), soapy water.
2. Press the "ON" button.
3. Insert the tip of the thermometer into the mouth, bottom, or under the armpit.
4. Hold the thermometer in place until it beeps (about 30 seconds).
5. Read the display.
6. Turn off the thermometer, rinse under water, and put it away in a safe place.

Caution

Alex wanted to measure the temperature of hot milk using a clinical thermometer. His teacher stopped him from doing so.

We are advised not to use a clinical thermometer for measuring the temperature of any object other than human body. Also we are advised to avoid keeping it in the sun or near a flame. Why?

A Clinical thermometer has small temperature range. The glass will crack/ burst due to excessive pressure created by expansion of mercury.



Maximum-minimum thermometer

The maximum and minimum temperatures of the previous day reported in weather reports are measured by a thermometer called the maximum - minimum thermometer.

1.5 Scales of thermometers

Celsius scale

Celsius is the common unit of measuring temperature, termed after Swedish astronomer, **Anders Celsius** in 1742, before that it was known as Centigrade as thermometers using this scale are calibrated from (Freezing point of water) 0°C to 100°C (boiling point of water). In Greek, '**Centium**' means 100 and '**Gradus**' means steps, both words make it **centigrade** and later **Celsius**.

Fahrenheit Scale

Fahrenheit is a Common unit to measure human body temperature. It is termed after the name of a German Physicist **Daniel Gabriel Fahrenheit**. Freezing point of water is taken as 32°F and boiling point 212°F. Thermometers with Fahrenheit scale are calibrated from 32°F to 212°F.

Kelvin scale

Kelvin scale is termed after **Lord Kelvin**. It is the SI unit of measuring temperature and written as K also known as absolute scale as it starts from absolute zero temperature.

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